Foundation for Virtual and Augmented Reality Systems

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Week – 01

Lecture – 02

Lec 2: Application Domains and System Categories

Hello and welcome to the NPTEL course on Foundations for Virtual and Augmented Reality Systems. We are going to start our second lecture, lecture number 2, where we will cover basically some application areas and categories of systems that we tend to see under virtual and augmented reality technologies. Before I start, let me just quickly recollect what we learned in the previous lecture.

In the previous lecture, we broadly introduced the field. Of course, many things were introduced at a very superficial level. As we progress in the course, we will slowly get into the details of those concepts. However, we got a broad overview of the field and also we got some introduction to the importance of the area, why we should know it, what is the potential, and so on.

Now, in this lecture, we are going to cover different application areas pertaining to extended reality, primarily virtual reality and augmented reality. And also, while discussing the application areas, we will also quickly have a look at the different categories of systems from the application point of view that are there in the market. Now, if you recollect from our previous lecture, we talked about some use forecasts, what is likely to be the scenario down the line in terms of application of this emerging technology that is extended reality. Of course, it covers both augmented reality and virtual reality. As we have seen, there is a huge potential in different application areas. Say for example, if we see the previous, previously what we have discussed,



see under video games, we have mentioned that the use forecast is that about 34 percent or more than one third of all the applications will include in one form or another the extended reality technologies. Then we have Healthcare, which again is going to have a significant component of these technologies as applications. Engineering, live events, video entertainment, real estate, retail, and military.

Now, sometimes these statistics can be misleading. Say, for example, under military, it is mentioned that 4% of the application areas are likely to include AR and VR technologies. which, of course, may change as things progress. But even if we consider 4%, that is a huge sum of money because military budgets worldwide are huge and this 4% translates to a significant amount of money.

So, both percentage-wise as well as value-wise, we can expect good penetration of these technologies in the commercial space. We have already discussed different market forecasts and other things in the previous lecture. Now, let us quickly have a look at the different applications that are currently there, which are coming up in different domains related to virtual reality. Then we will talk about augmented reality.

Now, again, if we recollect from the previous lecture, you can notice that we have used the term immersive technologies. So, VR technologies essentially refer to immersive technologies where our senses are blocked and we are transported to the virtual world. Now, there are several application areas pertaining to immersive technologies, several application domains are there and still it is evolving, So, newer domains may add. However, whatever is currently there, let us have a quick look.

The first thing is gaming and entertainment. This is where everything started, actually. As you can see, gaming and entertainment, of course, is a very old area. However, the recent

trend is that more and more gaming solutions and gaming products that are coming into the market rely on immersive technologies. So that it becomes more acceptable to the users and more widely popular.

So, this immersive gaming experience is a relatively new phenomenon, which is adding value to the gaming industry. And in this application, the gamer is expected to wear headsets or other VR devices to experience immersion while playing the game. Another prominent application domain for VR is healthcare. The applications primarily cater to medical education and training.

Now, here I would like to mention that not only VR, but MR is also used. What is MR? We will see in a later discussion. So, for example, here in this image as you can see the doctors can, potential doctors can learn about different body organs, how they work, how they look, what are the different parts in the organs while going through medical training.

Sometimes it is essential because there is a real problem of getting cadavers, and in such scenarios these simulations and immersive training modules can help a lot. Education is another emerging domain where immersive technologies have found lots of applications starting from the primary level to the tertiary level. So, it has been found that with an appropriate amount of exposure to immersive technologies can lead to improving learners' engagement and learning outcomes vis-a-vis the standard training, teaching, and learning methods.

But of course, we have to be careful while exposing a potential learner to the immersive environment; too long duration may create problems as we shall see later. As the image shows, this type of application caters to both primary education as well as tertiary education Classroom education is one prime example for primary level or secondary level education, whereas, higher educational institutes also come up with their own immersive educational solutions for different topic-wise teaching and learning. Although it is yet to gain too much popularity, it is an evolving area, that is the automobile industry.

So, typically what happens is that to increase customer satisfaction and customer experience, many companies are nowadays coming up with immersive experiences of using a newly launched car model without visiting, say for example, the showroom, or actually driving a car. Also, although it is primarily meant for potential buyers, it can also have other applications pertaining to the automobile industry including the design and development stage. Travel and tourism is also coming up heavily.

which includes these immersive technologies. The objectives are manifold. Say for example, if I want to give the experience of visiting a place to a user who cannot for some reason visit that place, that is called virtual tours, and virtual tourism has become quite popular nowadays. because it saves money, saves physical effort, while letting the user explore faraway places.

Another interesting thing that is coming up nowadays is the preservation of digital heritage and heritage tourism. So, for certain reasons, it is not possible to let a large number of users visit a heritage place, but or all parts of a heritage place from the point of view of preserving the heritage. But we can let the users visit the heritage place virtually, thereby enabling the user to explore each and every part of the heritage space without physically going there. Now that is one important application area that is evolving.

heritage tourism and this also leads to preservation of the heritage space. So, there can be physical decay, but we can preserve the space digitally in the form of heritage virtual tours, which can increase the lifespan and which can increase the potential of the space to be visited by many more users. Later on, we will talk about one case study where we will see how this digital heritage preservation is getting benefited by the use of these immersive technologies. Real estate and architecture, another prominent application area.

Now here, like automobiles, a potential buyer of a, say, upcoming real estate or upcoming property can explore the property without actually visiting the place, or it may so happen that the place does not even exist. It is yet to be made, but beforehand the user can experience and make an informed choice about whether to go for buying it or not. So, that is of course, of great benefit to potential buyers who wish to spend a lot of money on real estate.

Last but not least is defense applications. As I said, there is scope for several defense-related applications of immersive technology. Now there can be several use cases of application of such technologies. starting from specialized military training. So, here the actual vehicles, soldiers, and combat environment can be simulated in a virtual world with constraints put into place so that somebody can get trained on real-world scenarios.

Then this virtual boot camp and combat training can be another application area. where HMDs with motion trackers, full-body tracking systems, and wireless weapon controllers as per the user can be used for training. As we have already discussed, in medical education, immersive technologies play a crucial role in training. The same thing can be applied for military-related medical personnel. who wish to get trained on different scenarios.

So, we can have immersive technology applications for that. Even maintenance and assembly technician training can also be considered for such personnel in the military establishment. So, different forms of training including combat training, medical training, assembly line training, and so on, can be thought of. And there are already applications catering to these needs.

So, that is broadly what the major application areas of virtual reality technologies are. Now, there are many systems in those application areas and some other not-so-prominent application areas. Now, let us have a quick look at how these systems can be categorized. What are the major categories of systems?

This is from a systems point of view or systemic view of this field. So, we can roughly link it to the user experiences, what kind of experiences the user is going to get using those systems. Although at the beginning I mentioned that VR refers to immersive technology, the applications that are already developed that are in the market, they actually offer varying degrees of immersion. It is not that every application is fully immersive.

So, varying degrees of immersion can be expected from the applications that are already in the market, and depending on that degree of immersion we can categorize the VR systems. Now, as I said, depending on immersion, there can be categories. So, broadly there are three categories. First one is non-immersive.

This may sound counterintuitive because we are talking about immersive technologies whereas we are starting our system description with non-immersive. But why we can think of it as virtual reality from the system's point of view, we will discuss. The second category is semi-immersive. And the third category is fully immersive. Now, let us quickly have a look at each of these categories and what they actually refer to.

Non-immersive, as I said, seems to be counterintuitive. Why should it be part of virtual reality where we are talking about immersion? However, when we are referring to non-immersive VR systems, what we are referring to is virtual experience obtained through a computer where the user can control some characters or activities within the software but the environment is not directly interacting with the user. Now this definition, of course, is a bit cryptic, a bit complex, and we must admit that this is often overlooked as VR because already

These types of systems are very common. However, not all computer applications are considered to be non-immersive VR. We will be considering specific applications. Which

kind of applications? That provide computer-generated environments but allow users to keep control of their physical environment.

Now, these types of systems, They rely on computer or video game console displays, console displays, and input devices like keyboards and mice and also handheld controllers. Now, this seems to be our everyday use of computer systems. So, what is so special? Why are we calling it VR?

Before that, let us just quickly have a look at what kind of systems we are referring to. These are essentially gaming devices with the gaming experience such as PlayStation, Xbox, etc., which we are calling non-immersive VR. When we are playing a game with such systems such as World of Warcraft or similar games, we are calling it non-immersive. Note that we are specifically referring to games.

We are not saying any applications. Now why is that so? There can be other applications also which can be considered to be non-immersive if they satisfy the following. Now when I say immersion, essentially the idea is that our senses are blocked. So we are unable to receive any input from our real world environment.

And that is what immersion is. We are only getting input from the virtual world. Now, when we are actually playing a very engaging video game with video game consoles and accessories and input devices, many times it may be noticed that the user is fully quote unquote immersed in that game that means if you are saying something or something is happening outside the user is not aware of it although the eyes ears and other sensory organs are open but because the mind is completely in the game so they are oblivious to the real world surrounding

whatever is happening around them only if we are having such applications then we can call it non-immersive. So, it is kind of psychological engagement, although physically the user is attached to the real world. Gaming is a prime example. However, there may be other applications you can think of, which completely engage the user, completely immersing the user in the particular digital environment,

so that although other senses are open, they are still unable to experience it. That type of application we are calling non-immersive. This is purely a systemic point of view. Although psychologically that is the same as immersion. Now the next category of system is semi-immersive.

What is semi-immersive? A mixture of non-immersive and fully immersive VR. Ideally, we should not call it a mixture. We call it in between non-immersive and fully immersive VR. Where user can have partial,

Immersive experience by blocking some of the sensory organs, not all. Say if we are blocking only eyes, but other senses like our ears, our skin, they are exposed to real world signals, then we can call it partial immersion because only eyes are blocked. I cannot see anything; we cannot see anything from the real world. So, when we are using partially immersive systems, that means some of our senses are free to sense the real world around us. Now, it can be in the form of a 3D space or virtual environment where users can move on his or her own either through a computer screen or a VR box or headset, typically VR box.

A VR headset is the common example of a semi-immersive experience. And in such cases, users typically do not have any real physical movements. In most cases, in 90% of cases, you can say, other than the visual experience. Semi-immersive offers the most cost-effective and commonly used among all forms of VR systems. Virtual tours experienced with only wearing a headset can be an example of a semi-immersive system.

And this type of system can be both device-based as well as web-based, can be experienced by connecting to the web. Now, fully immersive, as opposed to non-immersive or semi-immersive, is a completely different class altogether. Now, this is actually what immersion means. That is, we are completely blocking the user's ability to sense the real world by blocking all the senses by some means. And the user is only sensing the things that are happening in the virtual world.

So, these types of systems are supposed to give a sense of being present fully in that virtual world and that everything is happening to the user for real. So, the user's sense of reality is replaced by virtual realism. As you can guess, these fully immersive systems are the costliest VR systems that we can have. They involve lots of devices to fully block our senses.

For example, helmets, gloves, body connectors with sense detectors, and body suits essentially. With those things on the body, users' movements, reactions, and even a blink of an eye can be detected and accordingly translated to actions in the virtual world. So, all these things are there to let the user feel or perceive that they are in the virtual world physically. So, they always perceive that they are in reality, whereas they are actually cut off from reality and in the virtual world.

So, if we are able to give the user that sense, then that type of system is called fully immersive. There are many examples we can have for virtual shooting game zones. In that zone, users will be wearing all these things; that is, a dedicated space where users will be wearing all these accessories. to experience a fully immersive virtual world. All users' actions will be translated to actions in the virtual world.

Users will not be able to sense anything in the real world. So different applications are there for these types of things. Say for example, users make actual movements; users make actual movements, move arms and body to run, jump, crouch, shoot, throw, and many more such things in the virtual world. User is viewing the environment through a helmet or head mounted display

and user is equipped with other gears to sense body movements and translate it to the things that are happening in the virtual world. Now, among the three categories of systems, fully immersive is the costliest one. Commercially, not many applications are available because it is difficult to afford. You will not find them in many places. More common are the semi-immersive and non-immersive systems.

However, non-immersive, as I said, are a special category of regular computer systems, software, which completely takes away your attention, completely immerses you in the interaction. Those are limited in set, limited in nature. More commonly, we can say that these are the semi-immersive ones. Now, semi-immersive ones give the feeling of immersion partially.

Of course, the user is still free to sense the real world. So most of the commercial immersive applications that you may come across are actually semi-immersive because they do not block all the sensors. So with that, we come to the end of this discussion on VR applications and system categories. What we have learned are the different application domains. We have covered a few important domains.

There may be other domains which are still evolving. but are yet to become very prominent. But as the use forecast says, VR is going to be a very important technology in multiple domains and a lot of effort will be given to develop such applications. The second thing is we talked about categories of VR applications. We mentioned three categories: non-immersive, semi-immersive, and fully immersive.

Now, that is about VR. So, in the next part of this lecture, we will be covering AR applications. So, with that, we come to the end of this part of the lecture. Hope you have understood the concepts and enjoyed the discussion.

See you again. Happy learning, and thank you.